



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



TOR VERGATA
UNIVERSITÀ DEGLI STUDI DI ROMA

Computational Fluid Dynamics

INTRODUCTION TO NUMERICAL METHODS FOR ASTROPHYSICS

Fritz Ali Agildere

January 23, 2026

Overview

Physical Background

- Fundamental Definitions

- Basic Equations

- Examples

Numerical Approaches

- Finite Difference Method

- Finite Volume Method

- Finite Element Method

Practical Implementation

- Kelvin-Helmholtz Instability

- Rayleigh-Taylor Instability

- Outlook

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- $M > 3$: Hypersonic (ram pressure stripping, dissociation and ionization)

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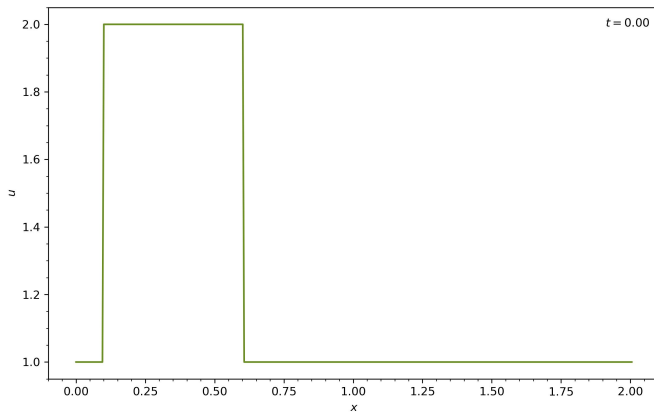
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Linear Advection

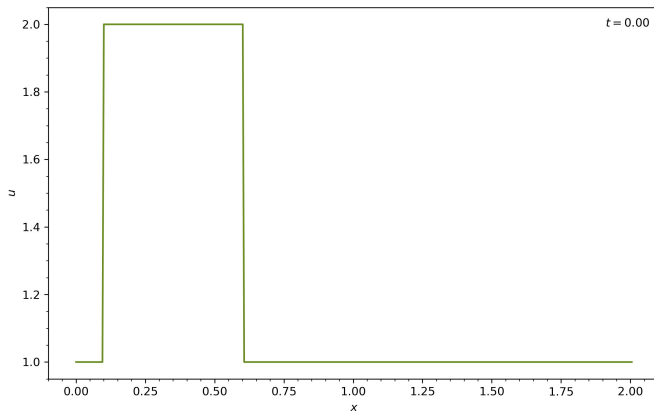
Linear Advection

■ *Initialization:*



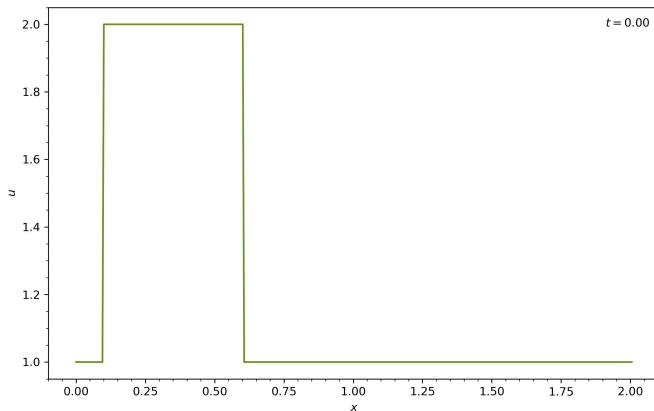
Linear Advection

- *Initialization:*
 - square wave



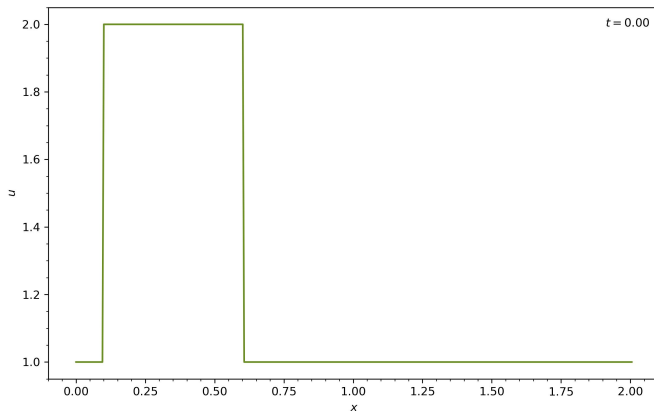
Linear Advection

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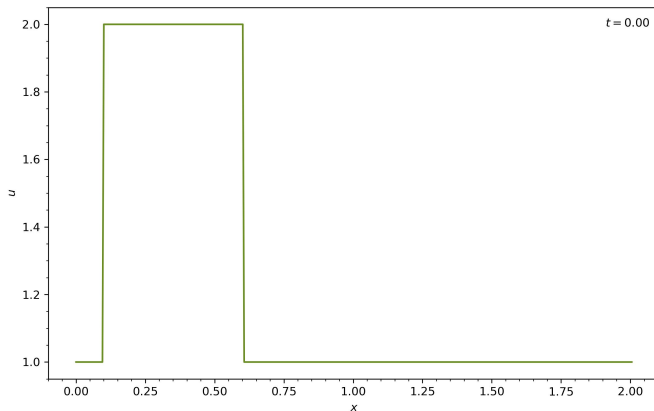
Linear Advection

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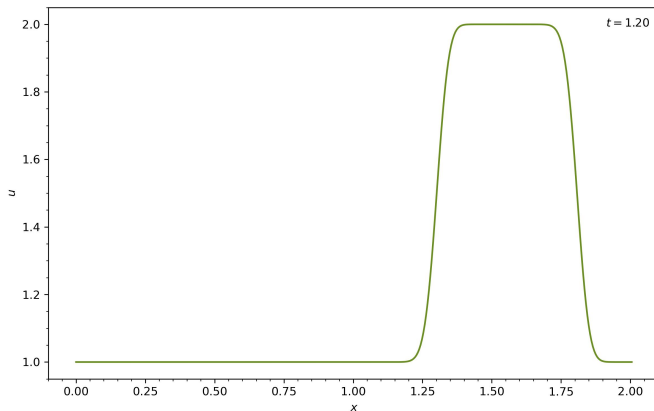
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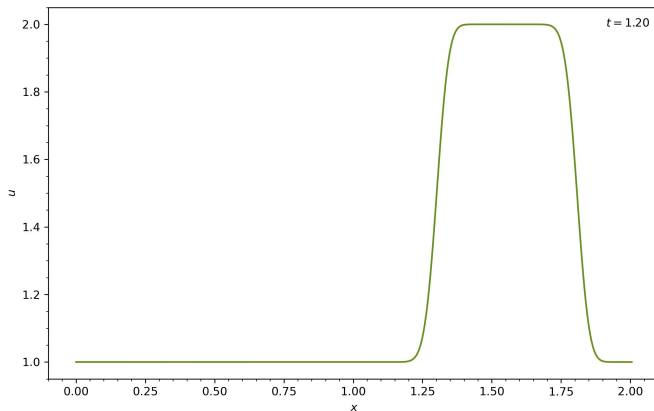
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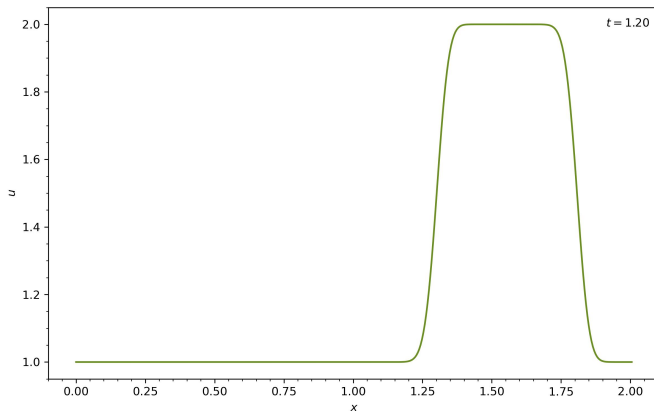
Linear Advection

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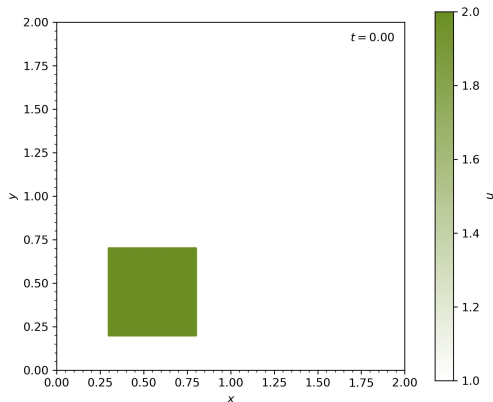
Linear Advection

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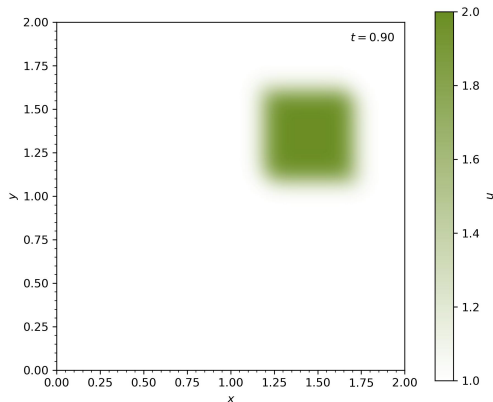
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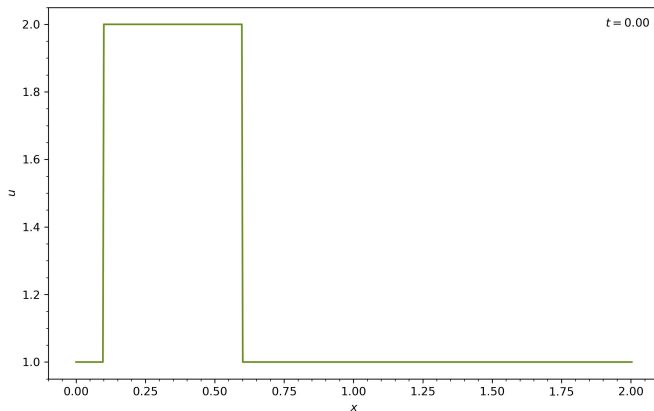
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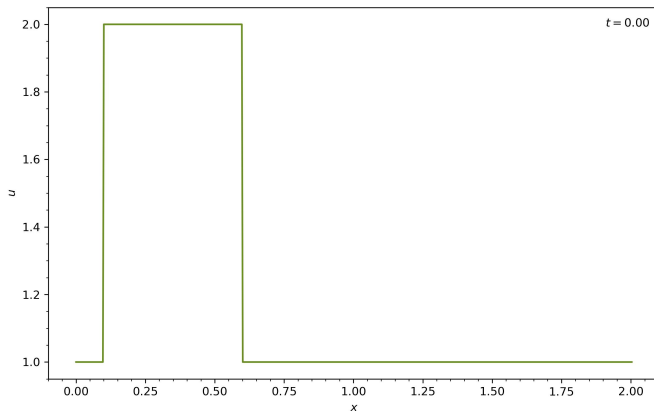
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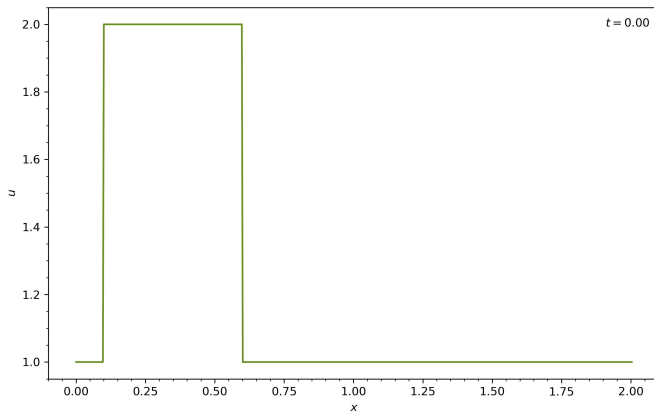
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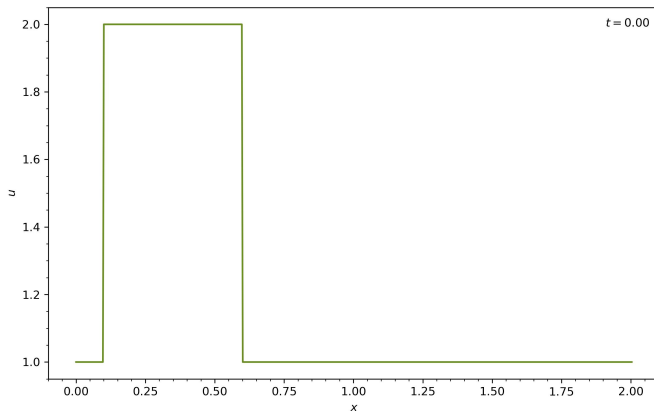
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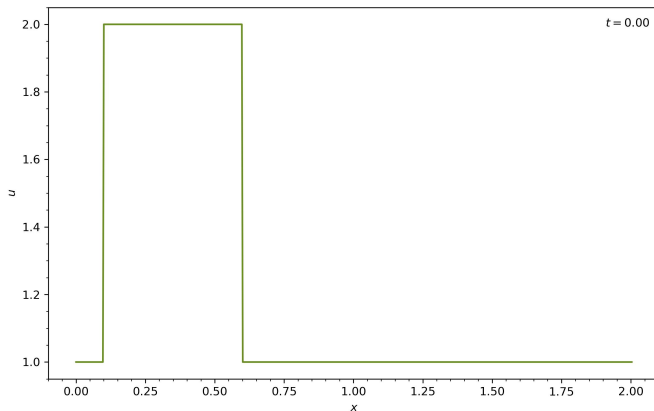
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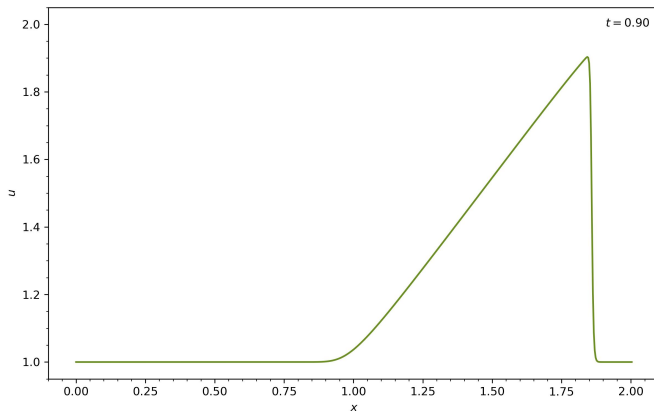
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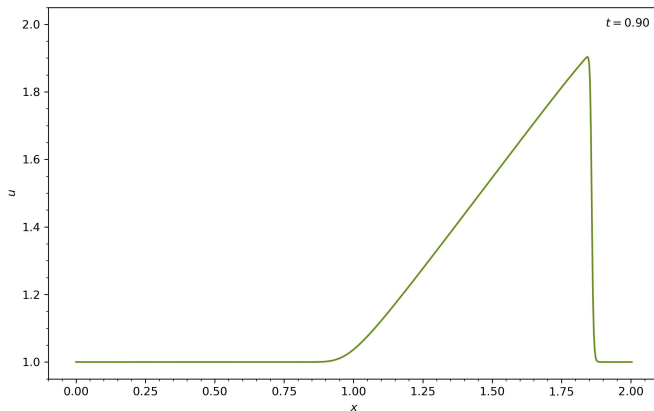
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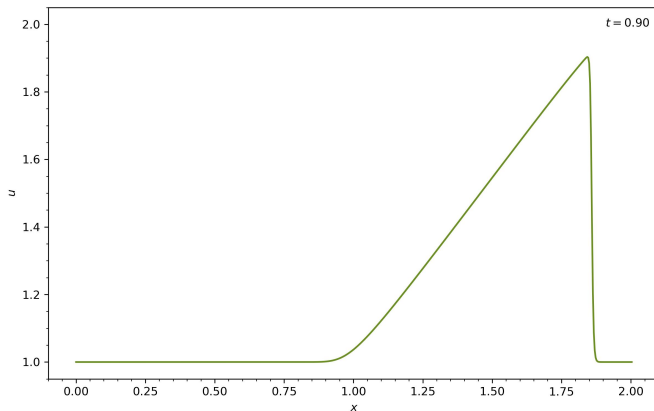
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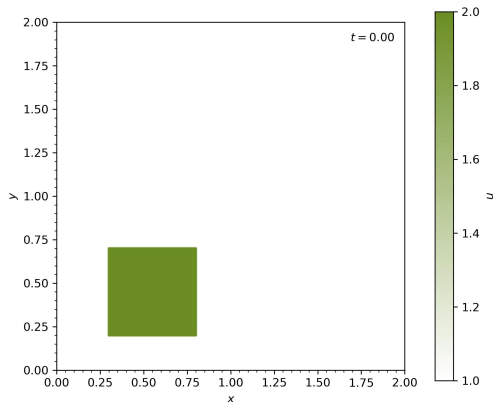
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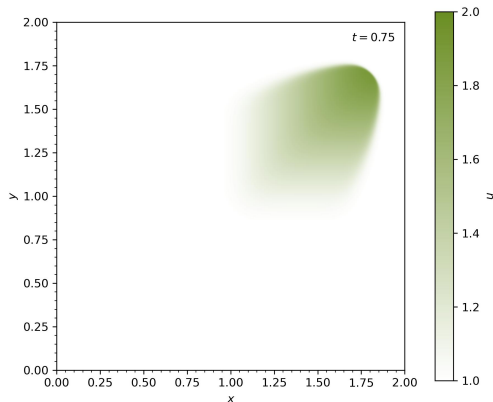
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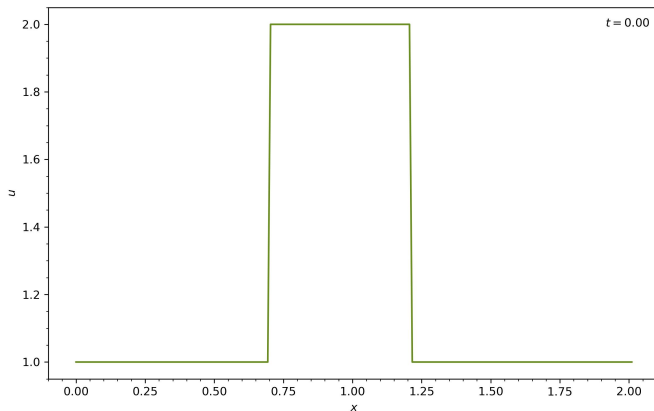
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Diffusion

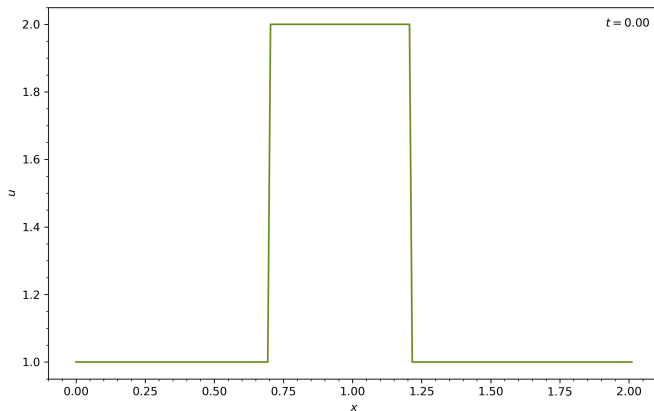
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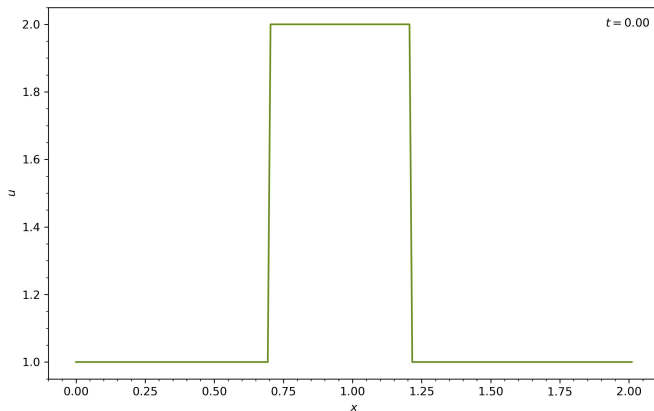
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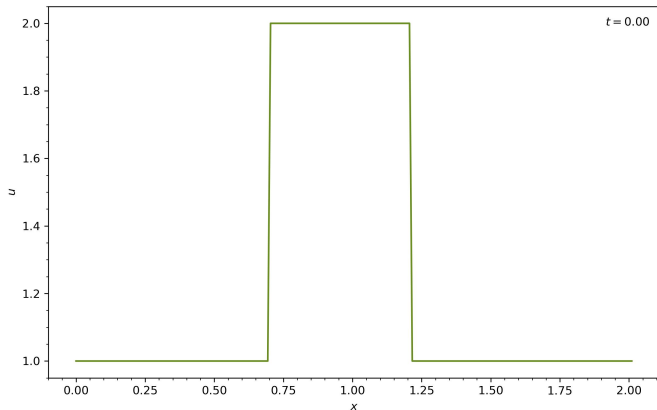
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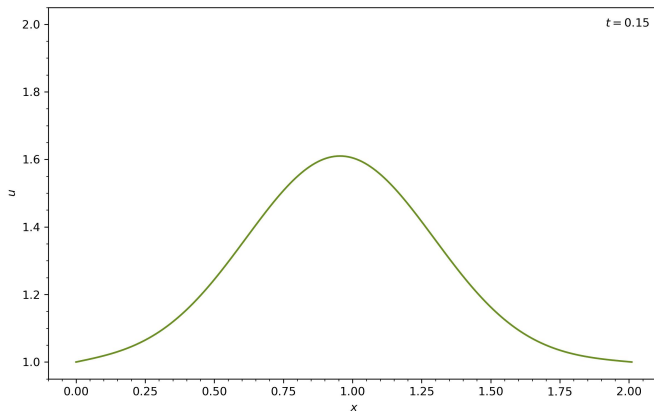
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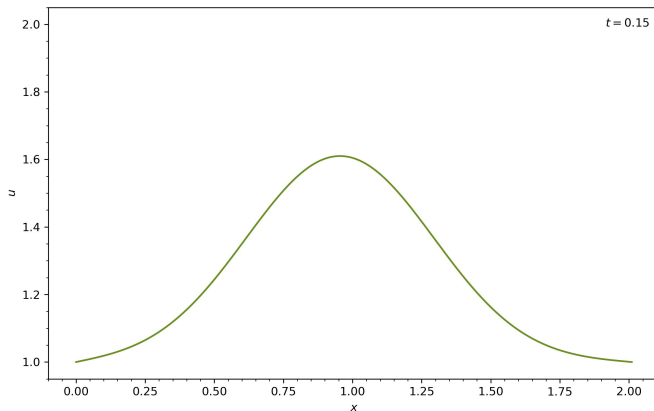
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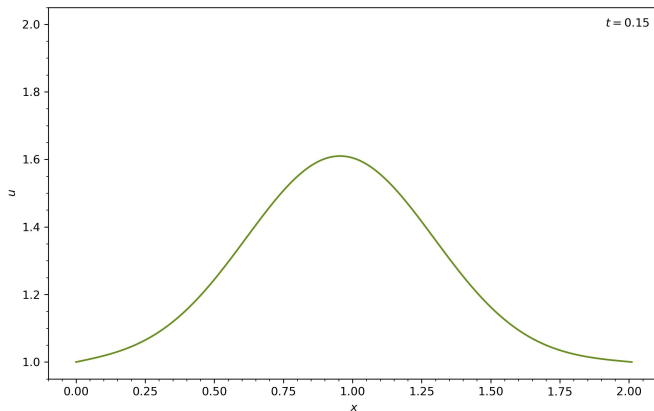
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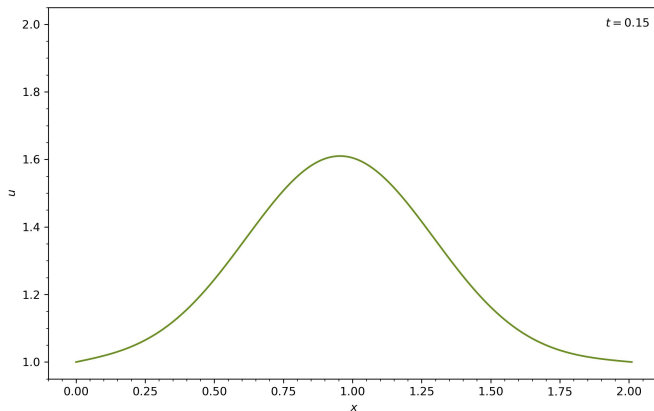
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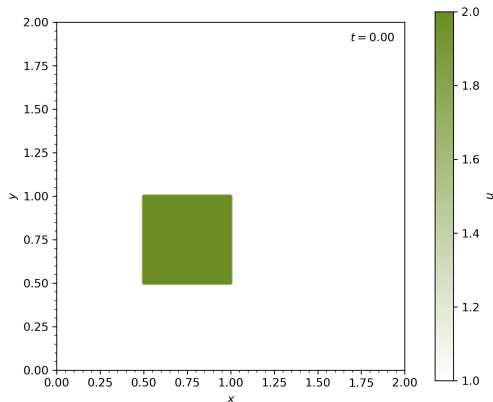
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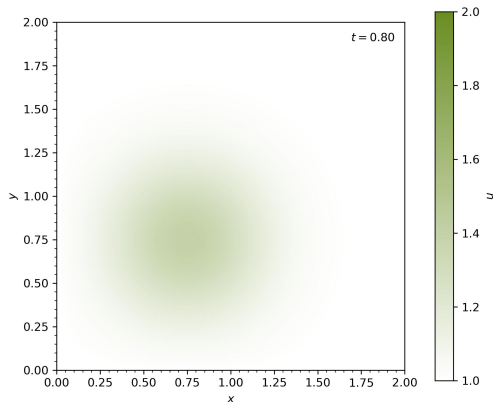
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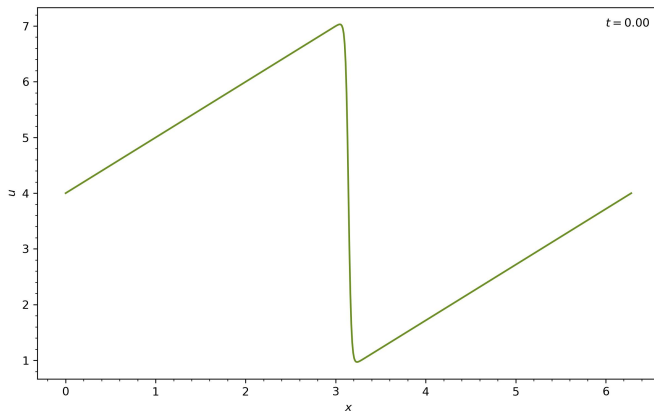
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Bateman-Burgers Equation

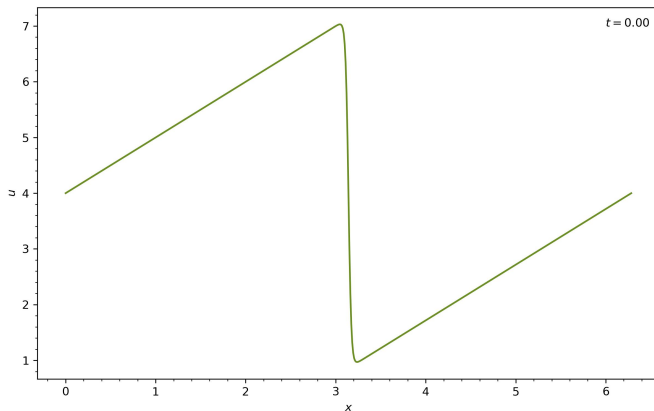
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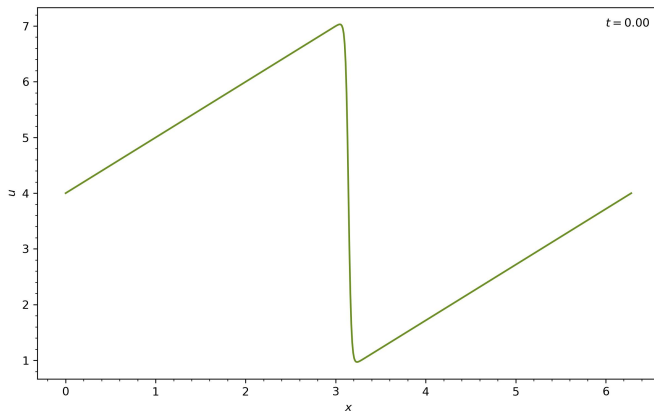
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- Initialization:
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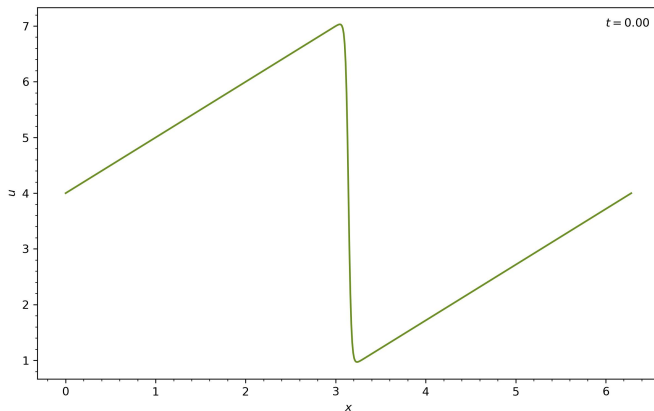
Bateman-Burgers Equation

- *Initialization:*
 - sawtooth wave
- *Boundaries:*



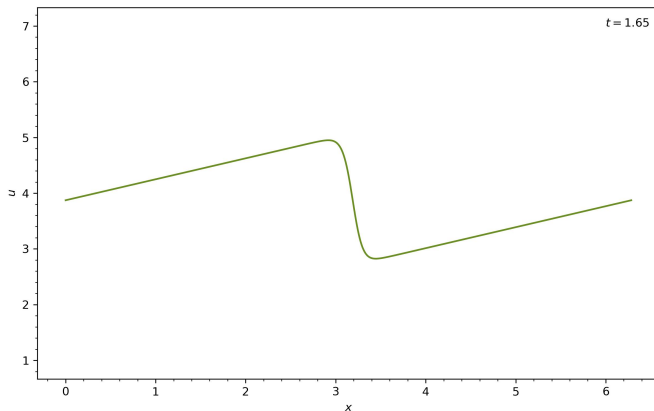
Bateman-Burgers Equation

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 - Both: periodic



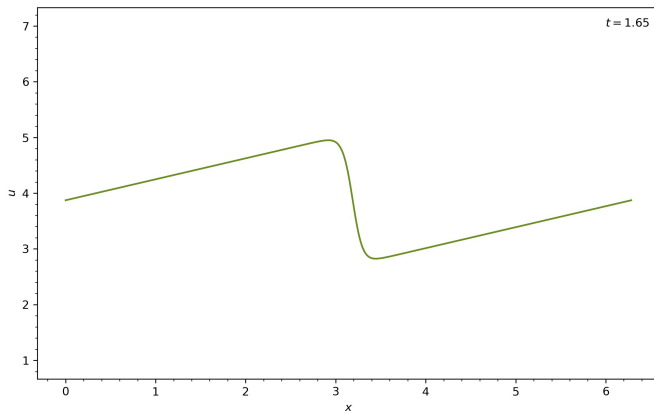
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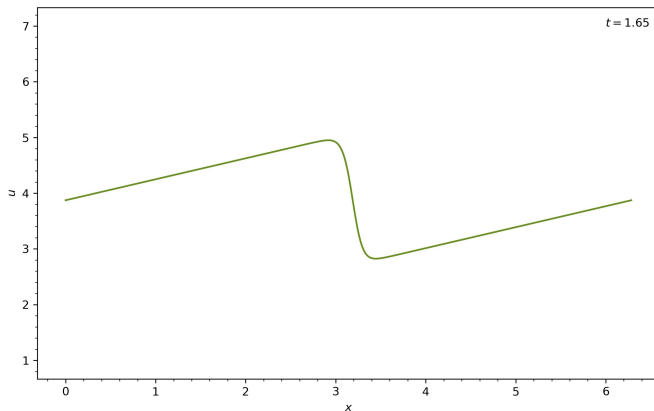
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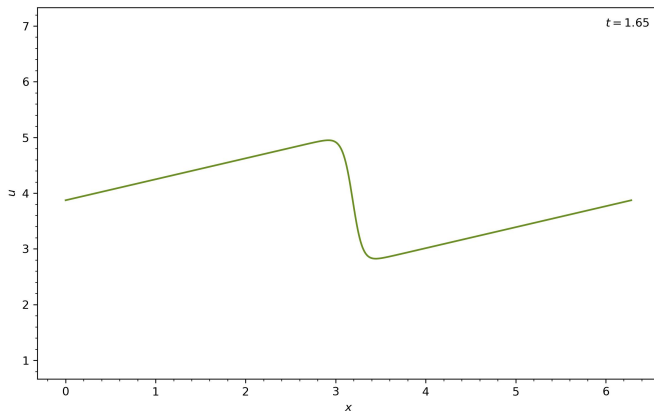
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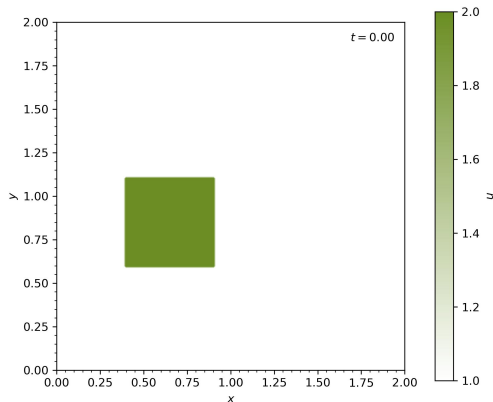
Bateman-Burgers Equation

- *Initialization:*
 - sawtooth wave
- *Boundaries:*
 - Both: periodic
- *Observation:*
 - nonlinear steepening
 - viscous smoothing
 - dissipating translation



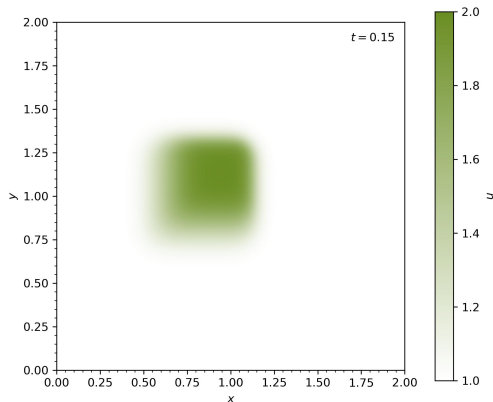
Bateman-Burgers Equation

- *Initialization:*
 - square wave
- *Boundaries:*
 - *Both:* constant value
- *Observation:*
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- Pressure: $\nabla^2 p = \rho \nabla \cdot ((u \cdot \nabla) u)$

Laplace Equation

Poisson Equation

Navier-Stokes Equations

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- fulfill at each step to accurately evolve initial configuration under given boundary conditions

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Lid Driven Cavity Flow

Pressure Driven Channel Flow

Nonuniform Periodic Flow

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Discretized Formulation

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- *Laplace Equation:* $\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \longrightarrow \frac{p_{i+1,j}^n - 2u_{i,j}^n + u_{i-1,j}^n}{\Delta x^2} + \frac{u_{i,j+1}^n - 2u_{i,j}^n + u_{i,j-1}^n}{\Delta y^2} = 0$

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- Idea: calculate flux through faces of control volumes that divide the domain using integral form

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 - bad stability for fast turbulent flows

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- *Adaptive Mesh Refinement*: grid resolution increases where higher precision is required

■ Regimes:

- | | |
|--|---|
| ■ <i>Plasma</i> : use magnetohydrodynamics | ■ <i>Degeneracy</i> : need quantum effects |
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■ Applications:

- stellar evolution
- accretion disks

Outlook

■ Frameworks:

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- | | | |
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| ■ stellar evolution | ■ accretion disks | ■ cosmic filaments |
|---------------------|-------------------|--------------------|

Thank You!

- C. Clarke & B. Carswell, *Principles of Astrophysical Fluid Dynamics* (Cambridge University Press) 2007
- L. A. Barba, *CFD Python: 12 Steps to Navier-Stokes*¹ 2022
- P. Mocz, *Medium*² 2020-2025

Resources on GitHub: [here](#)